Voter Turnout Prediction for Nevada, November 2014: Technical Report

Oct 10, 2026 · @omkar

1. Executive Summary

The model predicts that 37.5% of Nevada voters will turn out in November 2014, and it separates likely from unlikely voters well (validation AUC 0.92).

We trained a gradient boosting classifier on the 2010 general election, the most recent comparable midterm, using only information that existed before that election. Every feature is expressed as "elections ago", so the same rules apply when the model scores 2014 from 2012 and earlier history. The 2014 primary is excluded because it falls after the May 2014 cutoff.

On 20% of 2010 voters held back from training, the model reached 84.2% accuracy, 80.9% precision, 84.5% recall and an F1 score of 0.826. Voting in the previous general election, age, and voting two cycles earlier are the strongest drivers; profile fields such as net worth and ethnicity add little.

The overall 2014 rate should be read with care: general-election participation in the file falls from 74% (2002) to 26% (2012), so the probabilities are best used to rank voters.

2. Problem Definition

The task is to predict, for each voter in a Nevada voter file, whether they will vote in the November 2014 general election, pretending that today is May 2014.

The required output is, for every voter, a predicted outcome `vote` (1 = turns out, 0 = stays home) and a probability `vote_prob` between 0.00 and 1.00. Only information that would have been available by May 2014 may be used.

The file does not contain the 2014 general-election result, so the model cannot be trained on the outcome it must predict. We therefore learn from the closest earlier election with a known result (the 2010 midterm) and apply the learned pattern to 2014.

3. Dataset Description

The file `voterfile.csv` has 50,000 rows and 39 columns, but only 48,859 unique voters because 1,141 rows are exact duplicates.

The columns fall into four groups:

- **Identity and demographics:** `optimus_id`, `age`, `party`, `ethnicity`, `maritalstatus`, `dwellingtype`, `income`, `education`, `cd`, `dma`, `occupationindustry`.
- **Commercial and lifestyle fields:** `net_worth`, `home_owner_or_renter`, `petowner_dog`, `intrst_nascar_in_hh`, `intrst_musical_instruments_in_hh`, `donates_to_liberal_causes`, `donates_to_conservative_causes`.
- **Voter history (15 columns):** `vh00p` to `vh14p`. The digits are the election year, `p` is a primary and `g` a general election; 1 means the person voted and 0 means they did not. All 15 are complete and strictly 0 or 1.
- **Precinct turnout (6 columns):** general and primary turnout of each voter's precinct for 2008, 2010 and 2012.

Ages run from 19 to 100, with no values below 18 or above 100. `age` is missing for 17 rows.

4. Exploratory Data Analysis

The exploration shows falling general-election participation over time, very sparse lifestyle fields, and strong correlation between past votes.

Participation. Share of voters who voted in each general election: 47% (2000), 74% (2002), 51% (2004), 59% (2006), 36% (2008), 43% (2010) and 26% (2012). Primary participation is much lower, for example 17.8% (2008), 14.5% (2010), 11.8% (2012) and 8.75% for the 2014 primary. The decline toward 2012 is unusual, so a zero in the history may mean "did not vote" or "not yet registered".

Missing values. Five fields are more than 90% empty: `donates_to_liberal_causes` (99.8%), `donates_to_conservative_causes` (99.7%), `intrst_musical_instruments_in_hh` (99.1%), `intrst_nascar_in_hh` (95.5%) and `petowner_dog` (91.2%). Other gaps include `occupationindustry` (83.6%), `maritalstatus` (61.3%), `dwellingtype` (52.2%), `net_worth` (51.9%), `home_owner_or_renter` (47.6%), `education` (44.8%) and `ethnicity` (10.4%). `age` is missing for 0.03% of rows, and the voter-history columns have no missing values.

Precinct turnout. Average precinct general-election turnout was 0.62 (2008), 0.52 (2010) and 0.74 (2012); primary turnout averaged 0.13, 0.23 and 0.16. Each of these has only one missing value.

Other findings. The voter-history columns are correlated with each other, and the party mix is 19,900 Democratic, 17,299 Republican and 9,624 Non-Partisan, with smaller groups for other parties.

5. Data Preparation

Preparation removes duplicates, keeps ten profile fields, and turns the voter history into lag features that work for any target year. The original `voterfile.csv` is never modified.

- **Duplicates.** The 1,141 duplicated `optimus_id` values were all exact row copies (1,141 duplicate rows), so duplicates were dropped, leaving 48,859 unique voters. This also prevents the same voter appearing in both training and validation.
- **Exploration-only variables.** `historical_participation_count` (sum of all 15 history columns) and `age_group` were created during EDA. The count includes the 2014 primary, so neither is used as a model feature.
- **Dropped fields.** The five fields that are more than 90% empty and `occupationindustry` (83.6% missing) were left out.
- **Kept profile fields.** `party`, `ethnicity`, `income`, `education`, `maritalstatus`, `dwellingtype`, `net_worth`, `home_owner_or_renter`, `cd` and `dma`. Missing values are passed to the model as missing; `income` already has an "Unknown" level that stays as its own category. `cd` is a district label, so it is treated as categorical.
- **Lag features.** General and primary votes 2, 4, 6, 8 and 10 years before the target election (`gen_lag2` to `gen_lag10`, `pri_lag2` to `pri_lag10`), precinct general and primary turnout two years before (`prec_g_lag2`, `prec_p_lag2`), the counts `n_prior_general` and `n_prior_primary`, and `age_at_target`.
- **Eligibility.** For training, age is shifted back four years (age minus 4), and people younger than 18 in 2010 are excluded.

6. Temporal and Leakage Considerations

Every feature had to exist by May 2014, so the 2014 primary (`vh14p`) is excluded and no feature describes the election being predicted.

Variable	Decision	Reason
`vh00g` to `vh12g`, `vh00p` to `vh12p`	Used as lag features	Past elections, known by May 2014
`vh14p`	Excluded	The 2014 Nevada primary is held in June, after the cutoff (confirm the date and cite it)
`g08` / `p08` and `g12` / `p12` precinct turnout	Used as `prec_*_lag2`	Prior-election context

Variable	Decision	Reason
`g10` / `p10` precinct turnout	Not used in training	Describes the 2010 target election itself (target leakage)
`historical_participation_count`	EDA only	Includes `vh14p`
`optimus_id`	Not a feature	Identifier

Because the training target is 2010, its same-year primary (`vh10p`) is also left out, so training and scoring follow the same rule: no primary from the target year.

7. Predictive Method

The method is a gradient boosting classifier (scikit-learn `HistGradientBoostingClassifier`) with isotonic probability calibration and a 0.42 decision threshold.

Gradient boosting builds many small decision trees one after another, each correcting the errors of the previous ones. After random search tuning, the chosen settings were: `l2_regularization` 2.39, `learning_rate` 0.094, `max_depth` unlimited, `max_leaf_nodes` 15, `min_samples_leaf` 200 and `max_iter` 300. Isotonic calibration (5-fold) adjusts the raw scores so that `vote_prob` behaves like a real probability. A voter is predicted to vote (`vote` = 1) when `vote_prob` is at least 0.42.

A logistic regression with one-hot encoded categories served as the simpler benchmark.

8. Why the Method Was Selected

Gradient boosting was chosen because it scored best in cross-validation and suits this kind of data.

- **Better scores.** Five-fold cross-validation gave an AUC of 0.923 against 0.912 for logistic regression, and a lower Brier error (0.108 against 0.114).
- **Fits the data.** It handles missing values and categorical fields directly, and it can pick up interactions such as age combined with voting history without them being specified by hand.
- **Probabilities matter.** The assignment asks for a probability per voter, so a model whose output can be calibrated is a natural fit.
- **Explainable.** Permutation importance shows which features drive the predictions, which supports the campaign-team summary.

- **Honest comparison.** The gain over logistic regression is modest, which shows that even a simple, explainable model works well here.

9. How It Was Implemented

The work was done in Python (pandas, scikit-learn, matplotlib) in a Jupyter notebook, in nine steps:

1. Load `voterfile.csv`, run the EDA and data-quality checks, and remove the 1,141 duplicate rows.
2. Build lag features with a function that takes the target year, so the same code serves training (2010) and scoring (2014). For example, `gen_lag2` is `vh08g` in training and `vh12g` when scoring 2014.
3. Use `vh10g` as the training target, keep voters aged 18 or older at the target date, and split 80/20 with stratification (`random_state` 42).
4. Compare logistic regression and gradient boosting with 5-fold cross-validation on the 80% portion.
5. Tune gradient boosting with random search (25 candidates, 5 folds, 125 fits, scored by AUC).
6. Choose the threshold on out-of-fold predictions from the training portion only, searching 0.20 to 0.80 for the best F1 score. The result was 0.42.
7. Calibrate with isotonic regression and evaluate once on the held-out 20%.
8. Refit on all eligible 2010 voters and score every voter for 2014 using the same feature rules.
9. Save `predictions.csv` (`optimus_id`, the model features, `vote`, `vote_prob`) and validate it: unique IDs, `vote` only 0 or 1, `vote_prob` between 0 and 1, and no missing values in the three required columns.

Every data change in the notebook has a comment above it, as the assignment requires.

10. Model Evaluation

On 9,443 held-out 2010 voters the final model reached an AUC of 0.922 and 84.2% accuracy, and cross-validation and validation scores agree, so there is no sign of overfitting.

Cross-validation (training portion)

Model	AUC	Brier error	Log loss
Logistic regression	0.9124	0.1139	0.3705
Gradient boosting	0.9228	0.1081	0.3477

The best tuned model had a cross-validated AUC of 0.9227, so tuning did not improve on the defaults.

Held-out validation (threshold 0.42)

Model	AUC	Brier	Accuracy	Precision	Recall	F1
Original boosting	0.9214	0.1093	0.8414	0.8138	0.8355	0.8245
Tuned boosting	0.9215	0.1095	0.8419	0.8167	0.8324	0.8245
Tuned and calibrated (final)	0.9219	0.1094	0.8418	0.8091	0.8445	0.8264

The final confusion matrix has 4,392 true negatives, 839 false positives, 655 false negatives and 3,557 true positives. The three variants differ by less than the cross-validation noise (AUC standard deviation about 0.004), so tuning and calibration made only marginal differences; the final model was kept because it had the best AUC, recall and F1.

[Add here: the "voted in the previous general election" baseline result and the 2006 to 2010 forward backtest, once the notebook cells for them have been run.]

11. Results Interpretation

The model predicts that 18,332 of the 48,859 voters, or 37.5%, will vote in November 2014 (`vote` = 1), and the average `vote_prob` is 0.372.

Most voters receive a confident score: a quarter score below 0.065, the median is 0.169 and a quarter score above 0.832, with an overall range of 0.000 to 0.998. This split between clear "likely" and "unlikely" voters is what a campaign can use to prioritise contact.

Permutation importance (drop in AUC when a feature is shuffled) shows that past behavior and age dominate. Voting in the general election two years before the target (`gen_lag2`, 0.103) matters most, followed by age (0.086) and voting four years before (`gen_lag4`, 0.037). The primary two years before (0.013) comes next, while net worth, precinct turnout and ethnicity each contribute 0.003 or less. Party does not appear in the top ten. Permutation importance understates features that are correlated with one another, so this is a rough ranking.

The predicted rate of 37.5% is lower than the roughly 43% turnout observed for 2010 in the file, which is consistent with the downward trend in recent elections.

12. Limitations

The main limitation is that the 2014 predictions rest on a pattern learned from 2010, so the overall 2014 turnout rate is an estimate to be treated with caution.

- **No 2014 outcome to test against.** Validation used a random 20% of 2010 voters, which shows how well the model finds new voters in the same election, not how well it transfers to a different year.
- **Drifting base rate.** General-election participation in the file falls from 74% (2002) to 26% (2012). Probabilities are best used to rank voters, not as an exact count of who will vote.
- **Ambiguous zeros.** A 0 in the history may mean "did not vote" or "was not yet registered", and the file cannot tell these apart.
- **Midterm versus presidential years.** Only the 2010 midterm is used as the training analogue, so one election shapes the whole model.
- **Sparse commercial fields.** Many commercial fields are mostly empty, so demographic effects are hard to measure.
- **Importance is approximate.** Permutation importance understates features that move together.
- **Cutoff date.** The exclusion of `vh14p` relies on the 2014 Nevada primary date, which should be confirmed and cited.

13. Conclusion

A gradient boosting model trained on the 2010 midterm and applied to 2014 identifies likely voters accurately (AUC 0.92, accuracy 84%) and predicts that 37.5% of the 48,859 voters will turn out. Past voting and age explain most of the difference between voters and non-voters, so campaigns should rank contact lists by `vote_prob` rather than rely on demographics.

The four deliverables are:

1. **Predictions:** `predictions.csv` with `optimus_id`, the model features, `vote` and `vote_prob`.
2. **This report:** method, rationale, implementation and interpretation.
3. **Presentation:** a six-slide deck (one cover and five content slides).

4. **Code:** the complete Jupyter notebook, including data loading, checks, variable changes and comments.